

$$\begin{array}{ccccccccc}
 P_0 & \xrightarrow{f} & P_1 & \xrightarrow{g} & M & \longrightarrow & 0 & \longrightarrow & 0 \\
 \eta_{P_0} \downarrow & & \eta_{P_1} \downarrow & & \eta_M \downarrow & & \eta_0 \downarrow & & \eta_0 \downarrow \\
 T(P_0) & \xrightarrow{T(f)} & T(P_1) & \xrightarrow{T(g)} & T(M) & \longrightarrow & 0 & \longrightarrow & 0
 \end{array}$$